

LESSON 8-1 • SESSION 1 • PRACTICE • FORM B

Equations and Solutions

Learning target – I can check whether a value is a solution of an equation by substituting it. · 6.EE.B.5

REMEMBER

- An **equation** says two amounts are equal: $x + 3 = 10$.
- To test a value, **substitute** it for the variable. Then do the math.
- If both sides match, the value is a **solution**. $4k = 20$ with $k = 5$: $4 \times 5 = 20$, true.

WORKED EXAMPLE

A basket has b apples. You add 9 more and have 15. Is $b = 6$ a solution of $b + 9 = 15$?

Step 1 Substitute: replace b with 6 to get $6 + 9$.

Step 2 Add: $6 + 9 = 15$.

Step 3 Compare: $15 = 15$, so the equation is true.

Answer Yes, $b = 6$ is a solution.

WORDS equation (*ecuación*) · solution (*solución*) · substitute (*sustituir*)

PRACTICE

show your work · label units

1. Substitute the value. Write **T** for true or **F** for false.

a. $s - 12 = 30$ ($s = 42$) ____

b. $6r = 48$ ($r = 9$) ____

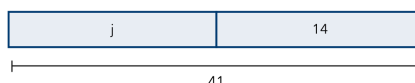
c. $p + 16 = 51$ ($p = 35$) ____

d. $3k = 27$ ($k = 24$) ____

e. $n - 5 = 4$ ($n = 9$) ____

f. $m + 19 = 44$ ($m = 25$) ____

2. **Juice Box Cooler.** A cooler holds j juice boxes. The coach adds 14 more. Now there are 41. The tape diagram shows $j + 14 = 41$.



a. Test $j = 27$. ____

b. Test $j = 25$. ____

c. Which value is the solution? ____

3. **Pencil Packs.** Tanisha wrote $8p = 104$ for packs of pencils. Find the solution. Prove it.

Show your work.

$p =$ _____

4. **Bake Sale.** Luis wrote $c - 18 = 47$ for cupcakes left. Is $c = 65$ a solution? Check.

Hint Replace c with 65. Do both sides match?

Show your work.

Yes or no _____

5. **Equal Teams.** Coach writes $5t = 35$ for equal teams. Ray says $t = 6$. Is Ray correct? Check.

Hint $5t$ means $5 \times t$.

Show your work.

Yes or no _____

6. **Library Fine.** Ana owes a library fine. After she pays \$3, she still owes \$14. Write an equation using f for the starting fine. Then find f by testing values.

Hint Start minus what she paid leaves what she owes.

Show your work.

Equation; f _____

7. **Game Tokens.** Deshawn wrote $7g = 56$ for the games he can play with 56 tokens. He says $g = 9$. Check his value. Then find the real solution.

Hint Substitute his value first. Then try the next number.

Show your work.

$g =$ _____

8. **Number Puzzle.** A puzzle reads: a number plus 19 equals 52.

- Write the equation using n . _____
- Test $n = 33$. Is it the solution? _____
- How is testing a value different from solving? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Devonte's Work.** To check whether $x = 15$ is a solution of $x + 25 = 45$, Devonte wrote $15 + 25 = 40$. He said that is close enough, so 15 is a solution. Explain his mistake.

TEST PRACTICE

choose the best answer

10. Is $x = 8$ a solution of $x + 7 = 15$?

A yes **B** no **C** only if $x = 7$ **D** cannot tell

11. Is $m = 9$ a solution of $4m = 32$?

A yes **B** no **C** only if $m = 4$ **D** cannot tell

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

13 A	55 Q	17 L	33 E	49 J	8 U	27 V	71 X	832 Z
		#2c	#3	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____